MAE 155A: Aerospace Engineering Design I

Weights and balance

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Overview

- Previously, we came up with an initial estimate for gross weight:
  1. Decomposed gross weight into empty, crew, payload, and fuel/battery weight.
  2. Used curve fits of historical data for empty weight fractions.
  3. Iteratively solved the resulting coupled equation.
  4. Looked at comparable values from previous similar aircraft.
- Now, we will decompose empty weight into the weights of individual aircraft components:
  - We will model the component weights as a function of component design parameters (e.g., area and aspect ratio for wings)
  - As before, the component weights will be computed using empirical curve fits.
  - Breaking the weight down by component enables predicting the location of the center of gravity (CG), which is important for stability analysis.

Weight build-up

Aircraft design textbooks provide component weight estimation formulas based on historical data.

Example—wing weight for GA aircraft (all imperial units):

$$W_{\text{wing}} = 0.036 S_w^{0.758} W_{fw}^{0.0035} \left(\frac{AR}{\cos^2 \Lambda} \right)^{0.6} q^{0.006} \lambda^{0.04} \left(\frac{100 t/c}{\cos \Lambda} \right)^{-0.3} (N_z W_{dg})^{0.49}$$

S_w - wing area

W_{fw} - fuel weight (carried in the wing)

AR - aspect ratio

Λ - quarter-chord sweep

λ - taper ratio

t/c - thickness-to-chord ratio

q - dynamic pressure

N_z - ultimate load factor, which is 1.5 times the limit load factor

W_{dg} - flight design gross weight

Weight build-up

Load factors

	$n_{positive}$	$n_{negative}$
General aviation—normal	2.5 to 3.8	−1 to −1.5
General aviation—utility	4.4	−1.8
General aviation—aerobatic	6	−3
Homebuilt	5	−2
Transport	3 to 4	−1 to −2
Strategic bomber	3	−1
Tactical bomber	4	−2
Fighter	6.5 to 9	−3 to −6

Category	Weight group	Fudge factor (multiplier)
Advanced composites	Wing	0.85–0.90
	Tails	0.83–0.88
	Fuselage/nacelle	0.90–0.95
	Landing gear	0.95–1.0
	Air induction system	0.85–0.90
Braced wing	Wing	0.82
Braced biplane	Wing	0.6
Wood fuselage	Fuselage	1.60
Steel tube fuselage	Fuselage	1.80
Flying boat hull	Fuselage	1.25
Carrier-based aircraft	Fuselage and landing gear	1.2–1.3

Fudge factors: For new configurations, new technologies, or concepts for which there is no historical data available

Raymer, D. (2012). Aircraft design: a conceptual approach. American Institute of Aeronautics and Astronautics, Inc..

Computing the CG

Aircraft components: wing, horizontal stabilizer, vertical stabilizer, fuselage, main landing gear, nose landing gear, engines, fuel system, flight controls, hydraulics, electrical, avionics, air conditioning, anti-ice, furnishings

Procedure:

1. Pick any reference point.
2. Estimate the weight of each component.
3. Select the CG position (x_{cg_i}) of each component in the aircraft frame, relative to the reference point.
4. Compute the aircraft CG relative to the reference point using the formula below.

The moment about the reference point is: $Wx_{cg} = \sum_i W_i x_{cg_i}$

Therefore, we can compute the CG position using: $x_{cg} = \dfrac{\sum_i W_i x_{cgi}}{\sum_i W_i}$

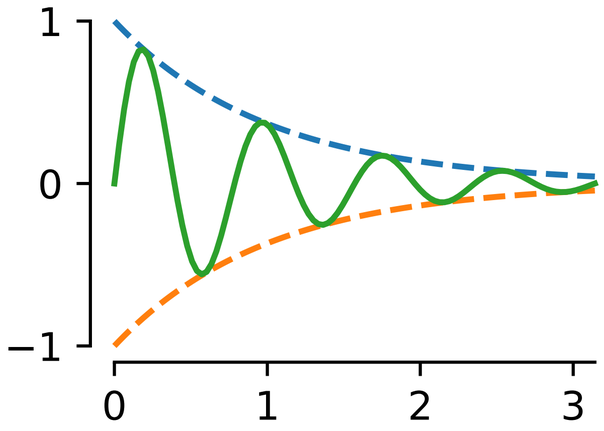
Static stability overview

Here, we are focused on trim and static longitudinal stability.

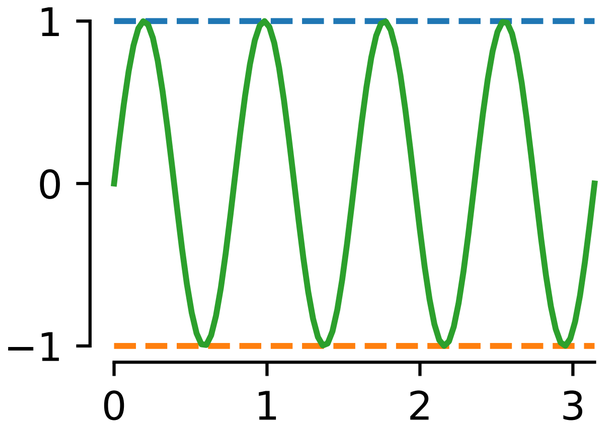
Trim: Balance of the forces and moments acting on the aircraft.

Static longitudinal stability: Considers whether pitching moments have a tendency to return to equilibrium given disturbances in angle of attack.

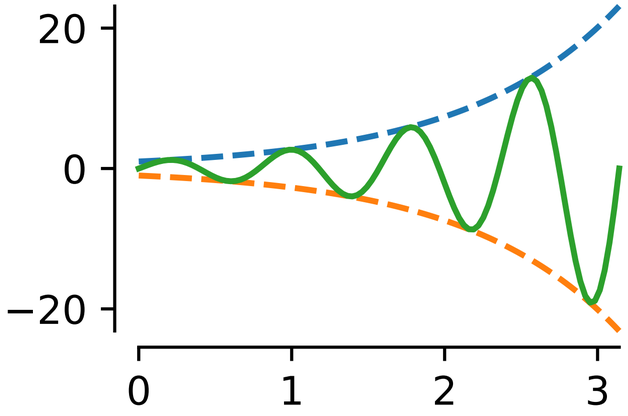
Dynamic stability: Considers the behavior of the aircraft over time in response to a disturbance from a steady flight condition.



Dynamically stable



Neutrally stable dynamically

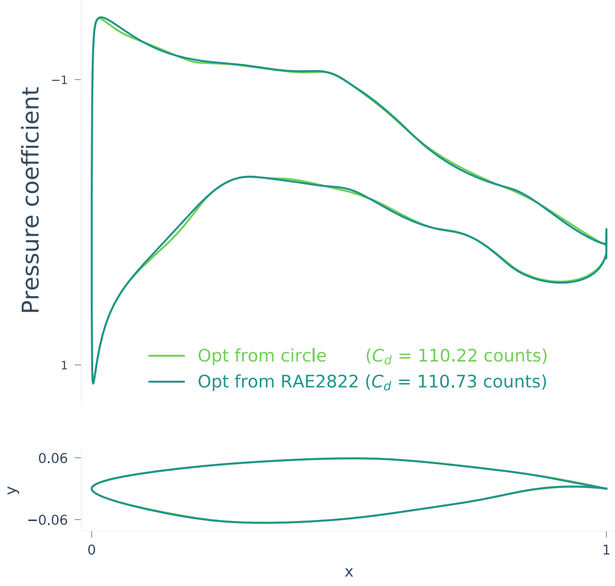


Dynamically unstable

Center of pressure and aerodynamic center

Pressure coefficient

$$C_p = \frac{p - p_\infty}{\frac{1}{2}\rho_\infty V_\infty^2}$$



He, X., Li, J., Mader, C. A., Yildirim, A., & Martins, J. R. (2019). Robust aerodynamic shape optimization—from a circle to an airfoil. Aerospace Science and Technology, 87, 48-61.

Center of pressure (CP): Point about which $C_m = 0$.

- The CP is the point at which we can represent a pressure field as a single force with no moment, at a particular flight condition (e.g., angle of attack)
- At all other points, we would have to represent the pressure distribution as a force at the point plus a non-zero moment.

Aerodynamic center (AC): Point about which $\dfrac{dC_m}{d\alpha} = 0$.

- The AC is the CP of the incremental pressure distribution that results from increasing the angle of attack.
- Since the AC is not the CP in general, the moment is non-zero at the AC.

$$x_{ac} = \frac{x_c}{4} + \Delta x_{ac}\sqrt{S_{wing}}$$

$$\Delta x_{ac} = 0.26(M - 0.4)^{2.5}, \quad (0.4 \leq M < 1.1)$$

$$\Delta x_{ac} = 0.112 - 0.004M, \quad (M > 1.1)$$

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Neutral point

Neutral point (NP): This is the AC of the entire aircraft.

The total moment acting on the aircraft is given by $C_m qSc = \sum_i \left(C_{m_i} qS_i c_i - C_{L_i} qS_i (x_{ac_i} - x_{ref}) \right)$

Dividing through by qSc, the total moment coefficient becomes:

$$C_m = \sum_i \left(C_{m_i} \frac{S_i}{S} \frac{c_i}{c} - C_{L_i} \frac{S_i}{S} \frac{x_{ac_i} - x_{ref}}{c} \right)$$

Taking the derivative with respect to α with the NP as the reference point, we obtain:

$$\frac{dC_m}{d\alpha} = \sum_i - \frac{dC_{L_i}}{d\alpha} \frac{S_i}{S} \frac{x_{ac_i} - x_{np}}{c} = 0, \text{ since } \frac{dC_{m_i}}{d\alpha} = 0.$$

Therefore, the neutral point is given by: $x_{np} = \dfrac{\sum_i \frac{dC_{L_i}}{d\alpha} S_i x_{ac_i}}{\sum_i \frac{dC_{L_i}}{d\alpha} S_i}$.

Tail lift-curve slope

$$x_{np} = \frac{\sum_i \frac{dC_{L_i}}{d\alpha} S_i x_{ac_i}}{\sum_i \frac{dC_{L_i}}{d\alpha} S_i}$$

Lifting surfaces downstream of others have modified lift-curve slopes because of downwash. Consider the horizontal tail:

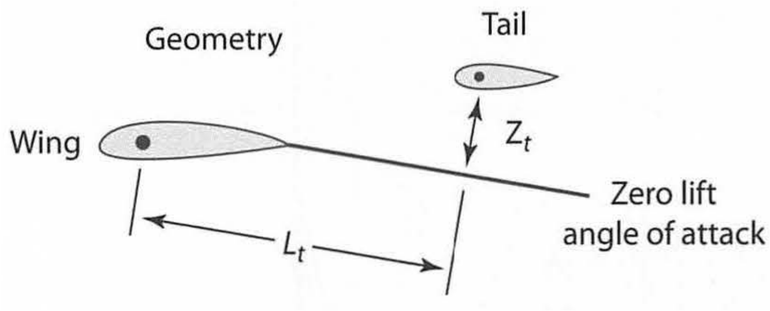
$$C_{L_t} = \left[C_{L_{t0}} + \frac{dC_{L_t}}{d\alpha_t}(\alpha + i_t - \epsilon) \right] \eta_h$$

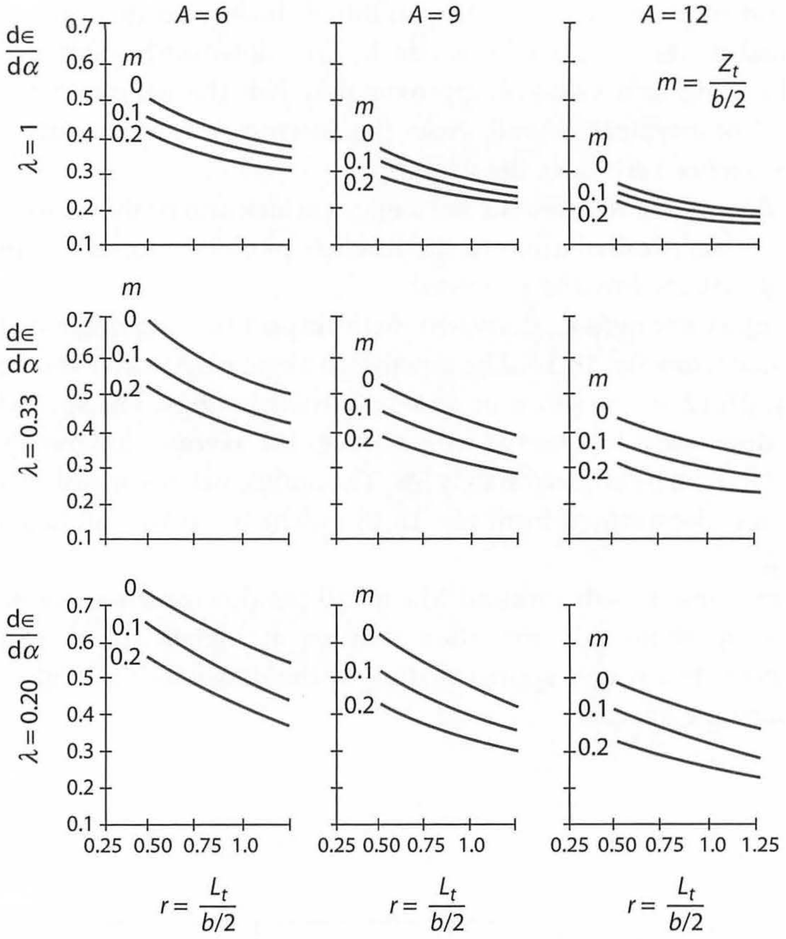
For the tail efficiency factor $\eta_h = \frac{q_h}{q}$, 0.90 is a typical value.

Taking the derivative with respect to angle of attack,

$$\frac{dC_{L_t}}{d\alpha} = \frac{dC_{L_t}}{d\alpha_t} \left(1 - \frac{d\epsilon}{d\alpha} \right) \eta_h$$

Tail downwash





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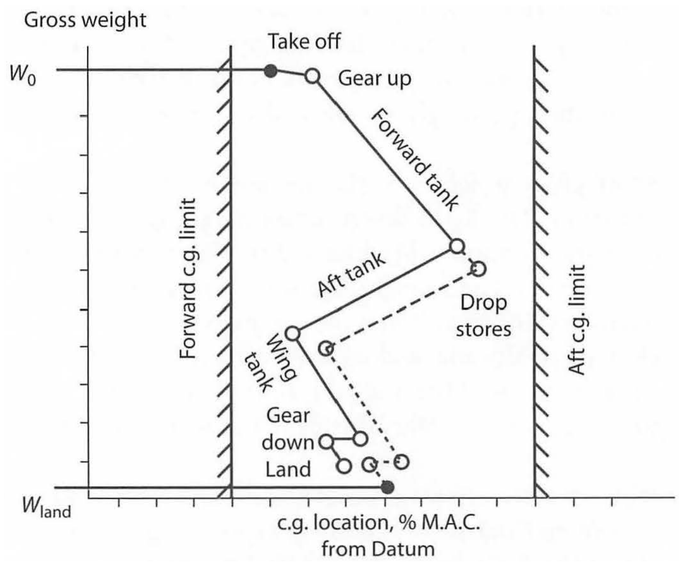
Static margin

$$SM = \frac{x_{np} - x_{cg}}{mac}$$

Typical values for GA aircraft are 10-20%.

CG excursion: Range of CG variation due to crew, fuel, and payload (passengers). For GA aircraft, the CG location varies up to 25%. For stability, we must maintain at least 5% static margin.



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Estimating tail sizing

Horizontal and vertical tail volume coefficients:

$$c_h = \frac{L_h S_h}{c_w S_w} \quad \text{and} \quad c_v = \frac{L_v S_v}{b_w S_w}$$

where the subscripts h and v stand for the horizontal and vertical stabilizer, respectively.

Note: L_* are the moment arms, measured from the quarter chord location of the wing to that of the corresponding stabilizer surface

	Typical values	
	Horizontal c_{HT}	Vertical c_{VT}
Sailplane	0.50	0.02
Homebuilt	0.50	0.04
General aviation—single engine	0.70	0.04
General aviation—twin engine	0.80	0.07
Agricultural	0.50	0.04
Twin turboprop	0.90	0.08
Flying boat	0.70	0.06
Jet trainer	0.70	0.06
Jet fighter	0.40	0.07–0.12*
Military cargo/bomber	1.00	0.08
Jet transport	1.00	0.09

Raymer, D. (2012). Aircraft design: a conceptual approach. American Institute of Aeronautics and Astronautics, Inc..